

GearDesign Report

Cylindrical gear(pair)

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Contents

1. Summary

- 1.1. Gear geometry
- 1.2. Total safety factor
- 1.3. Total damage & life
- 1.4. Others
- 1.5. Summary - Load cases

2. Detail geometry

- 2.1. Input gear geometry
- 2.2. Calculated geometry - Gear
- 2.3. Calculated geometry - Tooth
- 2.4. Calculated geometry - Mesh
- 2.5. Calculated geometry - Accuracy
- 2.6. Calculated geometry - Measurement

3. Detail rating

- 3.1. General
- 3.2. Pitting
- 3.3. Bending
- 3.4. Scuffing
- 3.5. Micro-pitting

4. Power loss

- 4.1. Mesh loss
- 4.2. Churning loss
- 4.3. Windage loss

1. Summary

1.1. Gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.5000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	10.0000	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	26.0000	57.0000
Facewidth (mm)	b	45.0000	45.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	105.3510	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Reference diameter (mm)	d	66.0030	144.6980
Tip diameter (mm)	d_a	71.003 (71.003 / 70.903)	149.698 (149.698 / 149.598)
Root diameter (mm)	d_f	59.753 (59.560 / 59.451)	138.448 (138.187 / 138.050)
Number of teeth spanned	k	4.0000	7.0000
Base tangent length (no backlash) (mm)	W_k	26.782 (26.716 / 26.679)	50.057 (49.968 / 49.921)
Effective Diameter of ball/pin (mm)	D_M	4.5000	4.2500
Diametral measurement over two balls (mm)	M_{dk}	72.641 (72.480 / 72.388)	150.469 (150.225 / 150.096)

1.2. Total safety factor

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]	Required S.F
Contact S.F	1.3722		1.4060		1.0000
Bending S.F	3.3278		3.4494		1.4000
Scuffing S.F (Flash Temp.)	3.4869				2.0000
Micropitting S.F	1.2573				1.0000

1.3. Total damage & life

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]
Contact Damage [%]	0.0000	0.0000	0.0000	0.0000
Bending Damage [%]	0.0000	0.0000	0.0000	0.0000
Contact Life [hr]	∞	∞	∞	∞
Bending Life [hr]	∞	∞	∞	∞
Total Req. Life [hr]	10000.0000			

1.4. Others

	Gear1 - Gear2
Transverse Contact Ratio	1.6033
Overlap Ratio	0.9949
Total Contact Ratio	2.5983
Worst Pitch Line Velocity [m/s]	34.5589
Total Efficiency [%]	97.9307
Worst Mesh Loss [W]	644.0762
Worst Churning Loss [W]	2459.9307
Worst Peak-to-Peak T.E. [μm]	0.1822
Gear Total Mass [kg]	2.2580
Hunting tooth	Complete

1.5. Summary - Load cases

[illegible]

2. Detail geometry

2.1. Input gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.5000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	10.0000	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	26.0000	57.0000
Facewidth (mm)	b	45.0000	45.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	105.3505	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Addendum coefficient	h^*_aP	1.0000	1.0000
Dedendum coefficient	h^*_fP	1.2500	1.2500
Root radius coefficient	ρ^*_fP	0.3800	0.3800
Tip form height coefficient	h^*_{FaP}	0.9600	0.9600
Ramp angle (deg)	α_{KP}	50.0000	50.0000
Protuberance height coefficient	h^*_{prP}	0.0000	0.0000
Protuberance angle (deg)	α_{prP}	0.0000	0.0000
Tip alteration (mm)	k_{m_n}	0.0000	0.0000
Tip radius coefficient (for topping)	ρ^*_{aP}	0.0000	0.0000
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Young's modulus (N/mm ²)	E	2.0600×10^5	2.0600×10^5
Poisson's ratio	ν	0.3000	0.3000
Density (kg/m ³)	ρ	7830.0000	7830.0000
Coeff. of thermal expansion (10 ⁻⁸ /°C)	α_{coeff}	11.5000	11.5000
Surface hardness	HRC	61.0000	61.0000
Core hardness	HBW	325.0000	325.0000
Tensile strength (N/mm ²)	σ_B	1200.0000	1200.0000
Yield strength (N/mm ²)	σ_s	850.0000	850.0000
Specific heat capacity (J/kg.K)	c_M	485.0000	485.0000
Specific heat conductivity (W/m.K)	λ_M	50.0000	50.0000
Mean roughness height, root (um)	R_{ZF}	20.0000	20.0000
Mean roughness height, flank (um)	R_{ZH}	4.8000	4.8000
Roughness average value, root (um)	R_{AF}	3.0000	3.0000
Roughness average value, flank (um)	R_{AH}	0.6000	0.6000

	Symbol	Pair1 Gear1	Pair1 Gear2
Fatigue strength for bending stress (N/mm ²)	σ_{Flim}	430.0000	430.0000
Fatigue strength for contact stress (N/mm ²)	σ_{Hlim}	1500.0000	1500.0000

2.2. Calculated geometry - Gear

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Gear ratio	GR	2.1923	
Transverse module (mm)	m_t	2.5386	
Working normal pressure angle (deg)	α_{wn}	20.0000	
Transverse pressure angle (deg)	α_t	20.2836	
Working transverse pressure angle (deg)	α_{wt}	20.2836	
Working transverse pressure angle, e/i (deg)	$\alpha_{wt}, e/i$	20.2997 / 20.2674	
Base helix angle (deg)	β_b	9.3913	
Helix angle at operating pitch circle (deg)	β_w	10.0000	
Radial backlash without tolerance (mm)	j_{r0}	0.0000	
Cumferential backlash without tolerance (mm)	j_{t0}	0.0000	
Min/max circumferential backlash by flank tolerance for each gear (mm)	$j_t, e/i$	0.0711 / 0.1117	0.0965 / 0.1472
Min/max normal backlash by flank tolerance for each gear (mm)	$j_n, e/i$	0.0658 / 0.1034	0.0893 / 0.1363
Cumferential backlash, e/i (mm)	$j_{tw}, e/i$	0.1594 / 0.2671	
Normal backlash, e/i (mm)	$j_{nw}, e/i$	0.1475 / 0.2471	
Angluar backlash, min (deg)	j_θ, min	0.2768	0.1262
Angluar backlash, max (deg)	j_θ, max	0.4637	0.2115
Effective facewidth (mm)	b_{eff}	45.0000	
Sum of the profile shift coefficients	Sum_x	-0.0000	
Generating profile shift coefficient	$x_E, e/i$	-0.0385 / -0.0604	-0.0522 / -0.0797
Prevent undercut profile shift coefficient	x_{Emin}	-0.5864	-2.4779
Max. profile shift coeff. (min top land)	x_{Emax}	1.2084	2.0702
Reference center distance, $x_1=x_2=0$ (mm)	a_d	105.3505	
Working center distance (mm)	a_w	105.3505	
Reference diameter (mm)	d	66.0027	144.6983
Operating pitch diameter (mm)	d_w	66.0027	144.6983
Operating pitch diameter with tolerance (mm)	$d_w, e/i$	66.0096 / 65.9958	144.7134 / 144.6832
27		71.0027	149.6983
Tip diameter with tolerance (mm)	$d_a, e/i$	71.0027 / 70.9027	149.6983 / 149.5983

	Symbol	Pair1 Gear1	Pair1 Gear2
Root diamter (mm)	d_f	59.7527	138.4483
Root diameter with tolerance (mm)	d_f, e/i	59.5604 / 59.4505	138.1873 / 138.0499
Tip form diameter (mm)	d_Fa	70.8027	149.4983
Tip form diameter with tolerance (mm)	d_Fa, e/i	70.5864 / 70.4632	149.2207 / 149.0749
Root form diamter (mm)	d_Ff	62.4849	140.3520
Root form diameter with tolerance (mm)	d_Ff, e/i	62.4123 / 62.3729	140.1621 / 140.0637
Active tip diameter (EAP) (mm)	d_Na	70.8027	149.4983
Active tip diameter with tolerance (mm)	d_Na, e/i	70.5864 / 70.4632	149.2207 / 149.0749
Active root diameter (SAP) (mm)	d_Nf	62.7717	141.1318
Active root diameter with tolerance (mm)	d_Nf, e/i	62.9591 / 62.8738	141.3445 / 141.2377
Mean diameter for mass calculation (mm)	d_m	65.2291	143.8834
Gear mass (kg)	m	0.6324	1.6255
Gear inertia (kg.m^2)	J	0.0000	0.0000
Nominal tip clearance (mm)	c	0.6250	0.6250
Effective tip clearance, e/i (mm)	c, e/i	0.8852 / 0.7445	0.8371 / 0.7102
Min. margin of involute contact, (d_Nf_i - d_Ff_e) / 2 (mm)	c_Nff	0.2308	0.5378

2.3. Calculated geometry - Tooth

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Tooth depth (mm)	h	5.6250	5.6250
Tooth depth, e/i (mm)	h, e/i	5.7761 / 5.6712	5.8242 / 5.7055
Working depth (mm)	h_w	5.0000	
Addendum (mm)	h_a	2.5000	2.5000
Addendum, e/i (mm)	h_a, e/i	2.5000 / 2.4500	2.5000 / 2.4500
Dedendum (mm)	h_f	3.1250	3.1250
Dedendum, e/i (mm)	h_f, e/i	3.2212 / 3.2761	3.2555 / 3.3242
Number of virtual gear teeth	z_n	27.1233	59.4626
Tooth thickness coeff.	s_nc	1.5708	1.5708
Transverse tooth thickness (mm)	s_t	3.9876	3.9876
Normal tooth thickness (mm)	s_n	3.9270	3.9270
Normal tooth thickness at d_a (mm)	s_an	1.8212	1.9636
Normal tooth thickness at d_a, e/i (mm)	s_an, e/i	1.7987 / 1.7031	1.9098 / 1.8137
Normal tooth thickness at d_Fa (mm)	s_Fan	1.9258	2.0522
Normal tooth thickness at d_Fa, e/i (mm)	s_Fan, e/i	2.0251 / 1.9197	2.1393 / 2.0244
Normal space width at d_f (mm)	e_fn	0.0000	1.9990
Normal space width at d_f, e/i (mm)	e_fn, e/i	0.0000 / 0.0000	2.0357 / 2.0560

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal pitch (mm)	p _n	7.8540	
Transverse pitch (mm)	p _t	7.9751	
Normal base pitch (mm)	p _{bn}	7.3803	
Transverse base pitch (mm)	p _{bt}	7.4806	
Axial pitch (mm)	p _x	45.2293	
Lead (mm)	p _z	1175.9612	2578.0688

2.4. Calculated geometry - Mesh

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Length of the path of contact (A to E) (mm)	g _α	11.9939	
Length of path of contact, e/i (mm)	g _α , e/i	11.4691 / 11.1012	
Length of approach POC of pinion (A to C) (mm)	g _{f1}	6.2573	
Length of approach POC of pinion, e/i (mm)	g _{f1} , e/i	5.9466 / 5.9030	
Length of recess POC of pinion (C to E) (mm)	g _{a1}	5.7366	
Length of recess POC of pinion, e/i (mm)	g _{a1} , e/i	5.5225 / 5.5026	
Length T1 - A (mm)	T1A	5.1831	
T1A tolerance (mm)	e/m/i	5.7232 / 5.6036 / 5.4839	
Length T1 - B (mm)	T1B	9.6965	
T1B tolerance (mm)	e/m/i	9.4724 / 9.4081 / 9.3438	
Length T1 - C (mm)	T1C	11.4405	
T1C tolerance (mm)	e/m/i	11.4504 / 11.4405 / 11.4305	
Length T1 - D (mm)	T1D	12.6637	
T1D tolerance (mm)	e/m/i	13.2038 / 13.0841 / 12.9645	
Length T1 - E (mm)	T1E	17.1770	
T1E tolerance (mm)	e/m/i	16.9530 / 16.8887 / 16.8244	
Length T2 - A (mm)	T2A	31.3384	
T2A tolerance (mm)	e/m/i	31.0058 / 30.9179 / 30.8300	
Length T2 - B (mm)	T2B	26.8250	
T2B tolerance (mm)	e/m/i	27.2094 / 27.1134 / 27.0173	
Length T2 - C (mm)	T2C	25.0810	
T2C tolerance (mm)	e/m/i	25.1028 / 25.0810 / 25.0592	
Length T2 - D (mm)	T2D	23.8578	
T2D tolerance (mm)	e/m/i	23.5252 / 23.4373 / 23.3494	
Length T2 - E (mm)	T2E	19.3445	
T2E tolerance (mm)	e/m/i	19.7288 / 19.6328 / 19.5367	
Length T1 - T2 (mm)	T1T2	36.5215	

	Symbol	Pair1 Gear1	Pair1 Gear2
T1T2 tolerance (mm)	e/m/i	36.5532 / 36.5215 / 36.4898	
Diameter of single contact point A (mm)	d_A	62.7717	149.4983
Diameter of single contact point B (mm)	d_B	64.8761	145.9442
Diameter of single contact point C (mm)	d_C	66.0027	144.6983
Diameter of single contact point D (mm)	d_D	66.8902	143.8685
Diameter of single contact point E (mm)	d_E	70.8027	141.1318
Transverse contact ratio	ϵ_α	1.6033	
Overlap ratio	ϵ_β	0.9949	
Total contact ratio	ϵ_γ	2.5983	
Total contact ratio with tolerance	e/m/i	2.528 / 2.504 / 2.479	
Specific sliding at Tip	ζ_a	0.4863	0.6374
Specific sliding at Root	ζ_f	-1.7579	-0.9467

2.5. Calculated geometry - Accuracy

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Quality Standard	-	ISO 1328: 2013	ISO 1328: 2013
Single pitch deviation (μm)	f_pt	8.5000	8.5000
Base circle pitch deviation (μm)	f_pb	8.0000	8.0000
Sector pitch deviation over k/8 pitches (μm)	F_pk/8T	17.0000	19.0000
Profile form deviation (μm)	f_fa	9.0000	9.0000
Profile slope deviation (μm)	f_Ha	7.0000	7.5000
Total profile deviation (μm)	F_α	11.0000	12.0000
Helix form deviation (μm)	f_fb	11.0000	11.0000
Helix slope deviation (μm)	f_Hβ	9.5000	10.0000
Total helix deviation (μm)	F_β	15.0000	15.0000
Cumulative pith deviation, total (μm)	F_p	26.0000	29.0000
Runout (μm)	F_r	23.0000	26.0000
Single flank composite, tooth-to-tooth (μm)	f_isT	8.5000	8.5000
Single flank composite, total (μm)	F_isT	34.5000	37.5000
Tooth-to-tooth radial composite deviation (μm)	f_idT	9.5000	9.5000
Total radial composite deviation (μm)	F_idT	31.0000	37.0000

2.6. Calculated geometry - Measurement

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Number of teeth spanned	k	4.0000	7.0000
Base tangent length (mm)	W_k	26.7822	50.0571
Base tangent length with tolerance (mm)	W_k, e/i	26.7164 / 26.6788	49.9678 / 49.9208
Tolerance of W_k (mm)	ΔW_k , e/i	-0.0658 / -0.1034	-0.0893 / -0.1363
Diameter of contact point (mm)	d_M	67.2801	144.3931
Theoretical diameter of ball/pin (mm)	D_M.theory	4.2730	4.2202
Effective Diameter of ball/pin (mm)	D_M.eff	4.5000	4.2500
Radial single ball measurement (mm)	M_rk	36.3205	75.2623
Radial single ball measurement with tolerance (mm)	M_rk, e/i	36.2402 / 36.1940	75.1403 / 75.0756
Diametral two ball measurement (mm)	M_dk	72.6410	150.4690
Diametral two ball measurement with tolerance (mm)	M_dk, e/i	72.4805 / 72.3879	150.2251 / 150.0958
Tolerance of M_dk (mm)	ΔM_{dk}	-0.1605 / -0.2530	-0.2438 / -0.3732

3. Detail rating

3.1. General

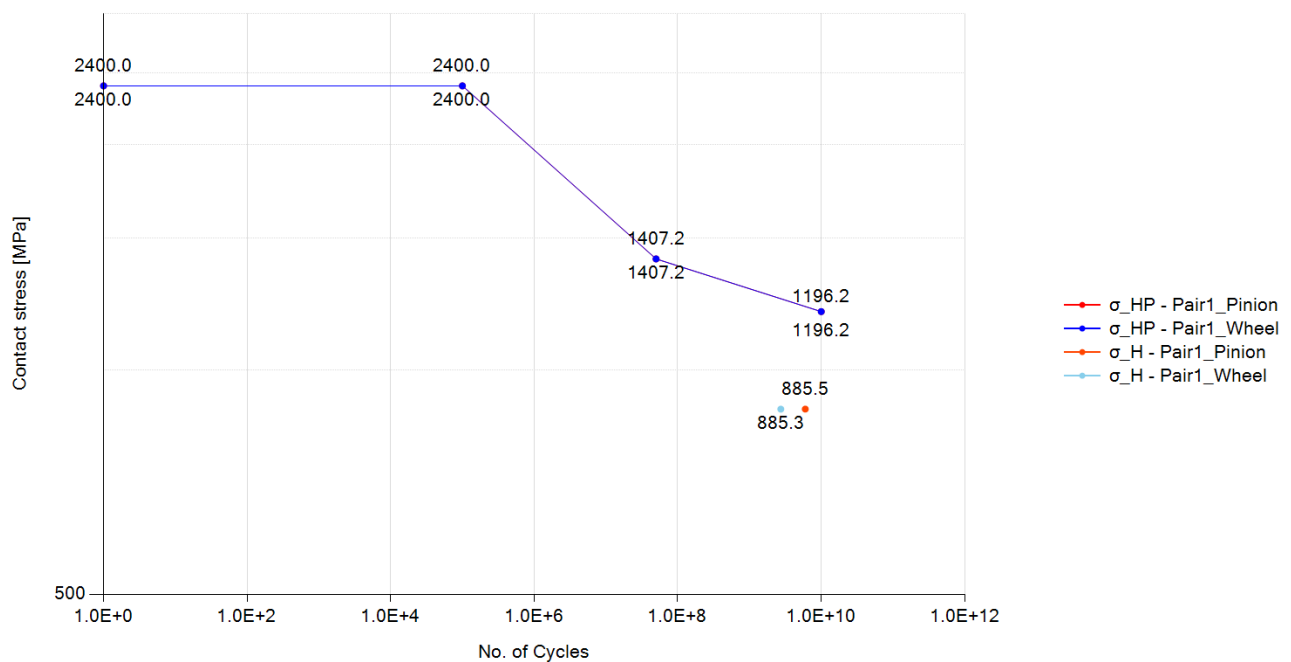
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rotational speed (rpm)	ω	10000.0000	4561.4040
Nominal tangential load at d (N)	F_t	4340.4100	
Nominal axial load at d (N)	F_a	765.3310	
Nominal radial load at d (N)	F_r	1604.1510	
Nominal normal load at d (N)	F_n	4690.2230	
Nominal tangential load per unit facewidth (N/mm)	w	96.4540	
Circumferential speed at d (m/s)	v	34.5590	
Sliding velocity at Tip dia. (m/s)	v_g	8.7480	9.5420
Nominal tangential load at d_w (N)	F_{tw}	4340.4100	
Nominal axial load at d_w (N)	F_{aw}	765.3310	
Nominal radial load at d_w (N)	F_{rw}	1604.1510	
Circumferential speed at d_w (m/s)	v_w	34.5590	
Running-in value (um)	y_p	0.6000	
Running-in value (um)	y_f	0.6750	
Effective single pitch & profile deviation (um)	f_{pb_eff}	7.4000	
Effective single pitch & profile deviation (um)	f_{fa_eff}	8.3250	
Correction factor	C_M	0.8000	
Gear blank factor	C_R	0.8980	
Basic rack factor	C_B	0.9750	
Material factor	E/E_{st}	1.0000	
Theoretical single stiffness (N/mm/um)	c'_{th}	17.4520	
Single tooth stiffness (N/mm/um)	c'	12.0380	
Mesh stiffness (for K_v , $KH\alpha$, $KF\alpha$), (N/mm/um)	$c_{y\alpha}$	17.4860	
Mesh stiffness (for $KH\beta$, $KF\beta$), (N/mm/um)	$c_{y\beta}$	14.8630	
Mean diameter for K_v (mm)	$d_m (K_V)$	65.3780	144.0730
Moment of inertia for K_v (kg.mm ²)	$J (K_V)$	581.5080	9174.0300
Moment of gear inertia per unit facewidth (kg.mm)	J^*	12.9220	203.8670
Relative gear mass per unit facewidth (kg/mm)	m^*	0.0135	0.0443
Relative gear mass per unit facewidth (kg/mm)	m_{red}	0.0103	
Resonance running speed (rpm)	n_{E1}	15105.7660	
Resonance ratio	N	0.6620	
Resonance ratio in main resonance ratio	N_s	0.8500	
Resonance range	-	subcritical range	
Dynamic factor	K_V	1.3540	

	Symbol	Pair1 Gear1	Pair1 Gear2
Face load factor - flank	$K_{H\beta}$	1.3000	
Face load factor - root	$K_{F\beta}$	1.2580	
Transverse load factor - flank	$K_{H\alpha}$	1.2070	
Transverse load factor - root	$K_{F\alpha}$	1.2070	
Number of load cycles	N_L	6.0000×10^9	2.7370×10^9

3.2. Pitting

S-N curve (Pitting), Load case 1, Meshing pair: Gear1 - Gear2



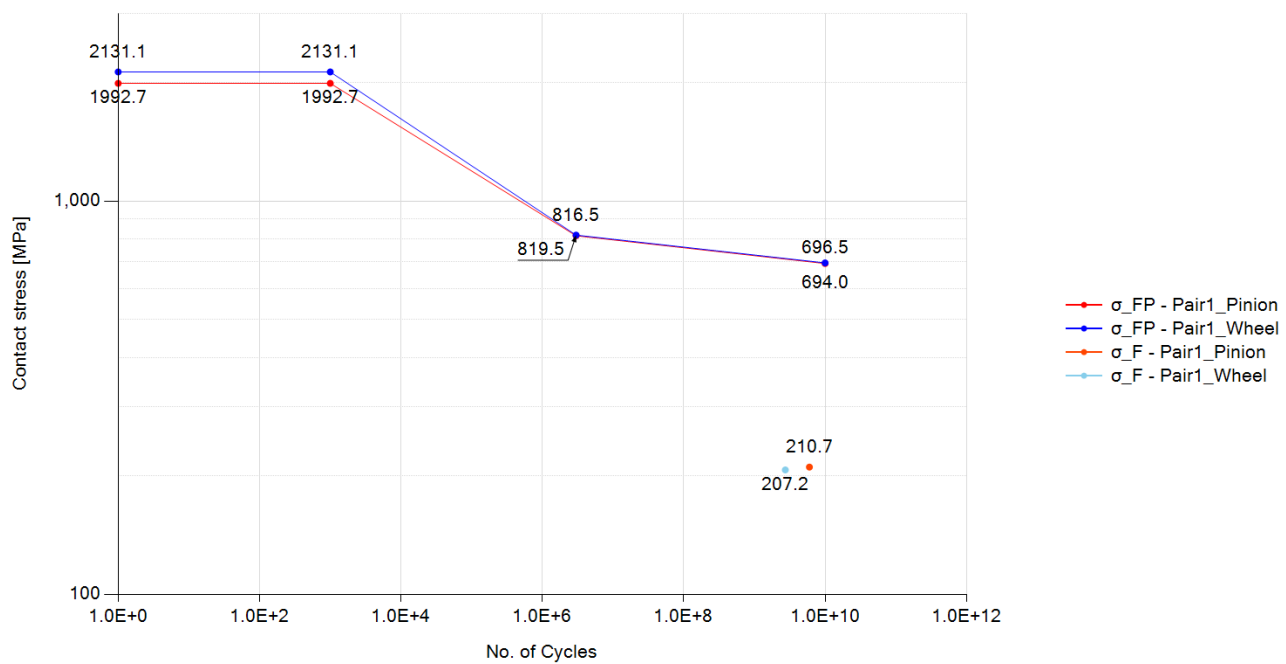
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-2:2019	
Contact ratio factor	Z_{ϵ}	0.7900	
Zone factor	Z_H	2.4630	
Elasticity factor	Z_E	189.8120	
Young's modulus for tooth flank (N/mm ²)	E	2.0600×10^5	2.0600×10^5
Helix angle factor	Z_{β}	1.0080	
Single pair tooth contact factor (pinion)	Z_B	1.0000	
Single pair tooth contact factor (wheel)	Z_D	1.0000	
Nominal contact stress at the pitch point (N/mm ²)	σ_{H0}	543.1920	
Contact stress (N/mm ²)	σ_H	885.5020	885.2760
Life factor	Z_{NT}	0.8630	0.8840

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubricant factor	Z_L	0.9460	0.9460
Velocity factor	Z_V	1.0370	1.0370
Roughness factor	Z_R	0.9570	0.9570
Work hardening factor	Z_W	1.0000	1.0000
Size factor	Z_X	1.0000	1.0000
Permissible contact stress - static (N/mm^2)	σ_{HP_static}	2400.0000	2400.0000
Permissible contact stress - mid point (N/mm^2)	σ_{HP_mid}	0.0000	0.0000
Permissible contact stress - reference (N/mm^2)	σ_{HP_ref}	1407.2430	1407.2430
Permissible contact stress - long (N/mm^2)	σ_{HP_long}	1196.1570	1196.1570
Permissible contact stress (N/mm^2)	σ_{HP}	1215.0470	1244.6570
Safety factor for Contact	S_H	1.3720	1.4060
Contact damage (%)	-	0.0030	0.0010
Cycle to failure	-	1.8100×10^{14}	1.8250×10^{14}

3.3. Bending

S-N curve (Bending), Load case 1, Meshing pair: Gear1 - Gear2



Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-3:2019	
Tooth form factor	Y_F	1.2070	1.1000
Stress correction factor	Y_S	1.8340	1.9800

	Symbol	Pair1 Gear1	Pair1 Gear2
Working angle (deg)	α_{Fen}	18.6450	19.8300
Bending moment arm (mm)	h_{Fe}	2.5830	2.7790
Tooth thickness at root (mm)	s_{Fn}	5.0260	5.4430
Tooth root radius (mm)	ρ_F	1.4400	1.3040
Helix angle factor	Y_{β}	0.9600	
Deep tooth factor	Y_{DT}	1.0000	
Rim thickness factor	Y_B	1.0000	1.0000
Nominal tooth root stress (N/mm ²)	σ_{F0}	81.9980	80.6520
Tooth root stress (N/mm ²)	σ_F	210.7000	207.2430
Life factor	Y_{NT}	0.8590	0.8720
Relative notch sensitive factor	$Y_{\delta relT}$	0.9920	0.9960
Surface factor	Y_{RrelT}	0.9570	0.9570
Size factor	Y_X	1.0000	1.0000
Stress correction factor	Y_{ST}	2.0000	
Mean stress influence factor	Y_M	1.0000	1.0000
Technical factor	Y_T	1.0000	1.0000
Permissible tooth root stress - static (N/mm ²)	σ_{FP_static}	1992.7000	2131.0650
Permissible tooth root stress - reference (N/mm ²)	σ_{FP_ref}	816.5110	819.4640
Permissible tooth root stress - long (N/mm ²)	σ_{FP_long}	694.0340	696.5440
Permissible tooth root stress (N/mm ²)	σ_{FP}	701.1740	714.8640
Safety factor for Bending	S_F	3.3280	3.4490
Bending damage (%)	-	0.0000	0.0000
Cycle to failure	-	3.5230×10^{28}	9.6320×10^{28}

3.4. Scuffing

Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubrication type	-	Spray lubrication	
Lubricant temperature (deg)	θ_{oil}	80.0000	
Lubricant viscosity (mPa.s)	η_{oil}	13.1510	
Load stage according to FZG A/8, 3/90	S_{FZG}	12.0000	
Gear density (kg/m ³)	ρ_M	7830.0000	7830.0000
Gear specific heat (J/kg.K)	c_M	485.0000	485.0000
Gear heat conductivity (N/s.K)	λ_M	50.0000	50.0000
Gear roughness (μm)	R_a	0.6000	0.6000
Number of mating gears	n_p	1.0000	
Effective facewidth (mm)	b_{eff}	38.2500	

	Symbol	Pair1 Gear1	Pair1 Gear2
Relevant tip relief (μm) (Min. value = 2 μm)	C_a	2.0000	2.0000
Transverse unit load (N/mm)	w_Bt	256.1940	
Lubrication system factor	X_S	1.2000	
Multiple meshing factor	X_mp	1.0000	
Structural factor	X_W	1.0000	1.0000
Structural factor (total)	X_Wtot	1.0000	
Thermal contact factor	B_M	13.7800	13.7800
Multiple path factor	K_mp	1.0000	
Optimal tip relief (μm)	C_eff	6.8950	
Angle factor	X_αβ	0.9760	
Lubricant factor	X_L	0.8790	
Roughness factor	X_R	0.7750	
Mean coefficient of friction	μ_m	0.0430	
Maximum flash temperature (deg)	θ_fl_max	67.5570	
Mean flash temperature (deg)	θ_fl_mean	25.0530	
Total mass temperature (deg)	θ_M	94.1300	
Scuffing temperature (deg)	θ_S	364.8370	
Maximum contact temperature (deg)	θ_Bmax	161.6870	
Safety factor for flash temperature	S_B	3.4870	

3.5. Micro-pitting

Load case 1

	Symbol	Gear1 - Gear2
minimum lubricant film thickness (μm)	h_Ymin	0.1592
minimum specific lubricant film thickness in the contact area	λ_GFmin	0.2654
permissible specific lubricant film thickness	λ_GFP	0.2111
safety factor against micropitting	S_λ	1.2573
transverse tangential load at reference cylinder per mesh (N)	F_t	4340.4100
transverse load in plane of action (N)	F_bt	4627.3600
reduced modulus of elasticity (N/mm ²)	E_r	2.2637×10 ⁵
thermal contact coefficient of pinion (N/(ms ^{0.5} *K))	B_M1	13779.6040
thermal contact coefficient of wheel (N/(ms ^{0.5} *K))	B_M2	13779.6040
failure load stage according to FVA 54/7 (90°)	-	0.0000
effective arithmetic mean roughness value (μm)	R_a	0.6000
roughness factor	X_R	1.1530
lubricant factor	X_L	1.0000
lubrication factor	X_S	1.2000

	Symbol	Gear1 - Gear2
elasticity factor (N/mm ²) ^{0.5}	Z_E	189.8120
helical load factor	K_Bγ	1.2400
load losses factor	H_v	0.1220
mean coefficient of friction	μ_m	0.0510
profile position at min. lubricant film thickness	-	0.0000
tip relief factor	X_Ca	1.0000
load sharing factor	X_A	0.6070
buttressing factor	X_but,A	1.2980
effective tip relief (μm)	C_eff	0.0000
Applied tip relief in calculation (μm)	C_a	0.0000
sum of tangential velocities at point A (m/s)	v_Σ,A	20.3970
sliding velocity at point A (m/s)	v_g,A	-9.5420
tangential velocity on pinion at point A (m/s)	v_r1,A	5.4280
tangential velocity on wheel at point A (m/s)	v_r2,A	14.9690
oil inlet/sump temperature (°C)	θ_oil	80.0000
bulk temperature (°C)	θ_M	95.9540
flash temperature at point A (°C)	θ_fl,A	79.1310
contact temperature at point A (°C)	θ_B,A	175.0860
pressure-viscosity coefficient at 38 °C (m ² /N)	α_38	1.8390×10 ⁻⁸
pressure-viscosity coefficient at bulk temperature (m ² /N)	α_θM	1.3600×10 ⁻⁸
pressure-viscosity coefficient at point A contact temperature (m ² /N)	α_θB,A	9.0560×10 ⁻⁹
kinematic viscosity at 40 °C (mm ² /s)	ν_40	68.0000
kinematic viscosity at 100 °C (mm ² /s)	ν_100	8.8000
kinematic viscosity at point A contact temperature (mm ² /s)	ν_θB,A	2.5030
dynamic viscosity at 38 °C (Ns/m ²)	η_38	0.0650
dynamic viscosity at bulk temperature (Ns/m ²)	η_θM	0.0080
dynamic viscosity at oil inlet/sump temperature (Ns/m ²)	η_θoil	0.0120
dynamic viscosity at point A contact temperature (Ns/m ²)	η_θB,A	0.0020
normal radius of relative curvature at point A (mm)	ρ_n,A	4.5080
material parameter	G_M	3078.3160
local velocity parameter at point A	U_A	8.0260×10 ⁻¹¹
local load parameter at point A	W_A	0.0002
local sliding parameter at point A	S_GF,A	0.1600
local nominal Hertzian contact stress at point A (Method B) (N/mm ²)	p_H,A	706.3170
local Hertzian contact stress at point A including the load factors K (Method B) (N/mm ²)	p_dyn,A	1151.1330

4. Power loss

4.1. Mesh loss

Load case 1

	Symbol	Gear1 - Gear2
Rating Method	-	ISO/TR 14179-2:2001
Input Power (kW)	P	150.0000
Gear ratio	u	2.1923
Tangential speed at working pitch dia. (m/s)	v_{wt}	34.5589
Tangential force at pitch dia. (N)	F_t	4340.4098
Tooth normal force, transverse section (N)	F_{bt}	4627.3596
Arithmetic average roughness of pinion	Ra1	0.6000
Arithmetic average roughness of wheel	Ra2	0.6000
Roughness factor	X_R	0.6000
Sum velocity (m/s)	v_{Σ}	23.9609
Equivalent radius of curvature (mm)	ρ	7.9634
Kinematic viscosity of oil at operating temp. (cSt)	η_{oil}	14.8602
Lubricant type	-	Mineral oil base
Lubricant factor	X_L	1.0000
distributed force on tooth surface(N/m)	F/b	150.0000
Mean coefficient of friction of the gear mesh	μ_{mz}	0.0352
Tooth loss factor	H_v	0.1221
Mesh power loss (W)	P_{VZP}	644.0762
Mesh efficiency (%)	η_z	99.5706

4.2. Churning loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	Changenet (2011)	Changenet (2011)
Rotating speed (rpm)	n	10000.0000	4561.4035
Lubricant	-	Oil: ISO-VG 68	Oil: ISO-VG 68
Gear immersed depth (mm)	h_e	33.0014	72.3491
Oil density (kg/m ³)	ρ	885.0000	885.0000
Operating Viscosity (cSt)	ν	14.8602	14.8602
dynamic viscosity (kg/m.s)	μ	0.0000	0.0000
Reynolds number	Re	1.0465×10^5	1.0465×10^5
Froude number	Fr	3689.0975	1682.7462
Centrifugal acceleration (m/s ²)	γ	16980.5614	4589.6848

	Symbol	Gear1	Gear2
Tip diameter of gear (mm)	d_a	71.0027	149.6983
Pitch diameter of gear (mm)	d_p	66.0027	144.6983
Facewidth (mm)	b	45.0000	45.0000
Dimensionless drag torque	C_m	0.0012	0.0028
Immersion area of the gear (m^2)	S_m	0.0152	0.0423
Total oil churning loss (W)	P	328.3697	2131.5610
Total oil churning torque (N.m)	T	0.3136	4.4624

4.3. Windage loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	None	None
Windage loss (W)	P_WL	0.0000	0.0000
Windage torque (N.m)	T_WL	0.0000	0.0000